DAMIEN CHEMISTRY OLYMPIAD NOTES

Chemical Bonding, Molecular Structure and Intermolecular Forces

Complete Question Bank: Questions Only

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A. Nature of Bonding

1. All Chemical Bonding as Electrostatic Attraction

1A Easy

Explain how the covalent bond in H_2 can be described entirely in terms of electrostatic attractions and repulsions.

1B Medium

Compare the electrostatic attractions responsible for bonding in NaCl, H_2 and magnesium metal. State what particles are attracting one another in each case.

1C Hard

A student says, “Ionic bonds are electrostatic, whereas covalent bonds arise from electron sharing rather than electrostatic forces.” Evaluate this statement, accounting for electron–nucleus, nucleus–nucleus and electron–electron interactions.

2. Ionic, Covalent, Coordinate and Metallic Bonding

2A Easy

Identify the principal type of bonding in Na_2O, Cl_2, NH_4^+ and solid copper.

2B Medium

Explain the formation of the coordinate bond in NH_4^+ . Why are the four N–H bonds indistinguishable after the ion has formed?

2C Hard

Describe all the important types of bonding present in $[Cu(NH_3)_4]SO_4$, distinguishing between bonding within the complex ion, within ammonia and between the ions in the solid.

3. Electron Transfer Versus the Resulting Ionic Attraction

3A Easy

Distinguish between the transfer of an electron from sodium to chlorine and the ionic bond present in solid NaCl.

3B Medium

Electron transfer from gaseous sodium to gaseous chlorine involves both endothermic and exothermic steps. Explain why the formation of solid NaCl can nevertheless be strongly exothermic.

3C Hard

A diagram shows one Na^+ ion beside one Cl^- ion and labels the pair “a molecule of sodium chloride”. Explain why this is a misleading representation of solid NaCl.

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4. Delocalisation of Electrons

4A Easy

What is meant by a delocalised electron? Give one metallic and one covalent example.

4B Medium

Explain why the two C–O bonds in the ethanoate ion have the same length even though a single Lewis structure shows one single bond and one double bond.

4C Hard

The three N–O bonds in nitrate have equal lengths intermediate between typical N–O and N=O bonds. Use electron delocalisation to predict the average N–O bond order.

5. Dot-and-Cross Diagrams

5A Easy

Draw dot-and-cross diagrams for Cl_2 and $MgCl_2$.

5B Medium

Draw a dot-and-cross diagram showing the formation of NH_4^+ from NH_3 and H^+ , clearly indicating the origin of the bonding electrons.

5C Hard

Gaseous $AlCl_3$ can dimerise to form Al_2Cl_6. Draw a suitable dot-and-cross representation and explain why coordinate bonds form during dimerisation.

6. Formal Charge

6A Easy

Calculate the formal charge on each atom in NH_4^+ .

6B Medium

Draw the major Lewis structures of the cyanate ion, OCN^- . Calculate the formal charges and identify the most important contributor.

6C Hard

Several Lewis structures can be proposed for N_2O with atom order N–N–O. Assign formal charges and decide which structures contribute most strongly to the resonance hybrid.

7. Resonance

7A Easy

Explain why carbonate is represented using several resonance structures rather than one fixed Lewis structure.

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7B Medium

Benzene is sometimes drawn with alternating single and double bonds. Explain why this does not mean that benzene rapidly switches between two different structures.

7C Hard

The C–N bond in an amide is shorter than a normal C–N single bond, and the atoms around nitrogen are approximately planar. Explain both observations using resonance.

B. Orbital Overlap

8. Formation of σ and π Bonds

8A Easy

State the number of σ and π bonds in ethane, ethene and ethyne.

8B Medium

Describe the orbital overlaps involved in forming the carbon–carbon bonds in ethene and ethyne.

8C Hard

Allene, $H_2C{=}C{=}CH_2$, contains two adjacent double bonds. Predict the relative orientations of its two terminal CH_2 groups using the orientations of the π orbitals.

9. Relative Strength of σ and π Overlap

9A Easy

Explain why a σ bond is normally stronger than a π bond formed from the same atomic orbitals.

9B Medium

A C=C bond is stronger than a C–C bond, but not twice as strong. Explain this observation.

9C Hard

Ethene undergoes addition reactions much more readily than ethane even though the total C=C bond is stronger than the C–C bond. Resolve this apparent contradiction.

10. Restricted Rotation Around Double Bonds

10A Easy

Explain why free rotation is possible around the C–C bond in ethane but not around the C=C bond in ethene.

10B Medium

Draw the geometrical isomers of but-2-ene and explain why they can exist as separate compounds.

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10C Hard

Rotation around the C–N bond in an amide is restricted even though it is usually drawn as a single bond. Explain this using orbital overlap and resonance.

11. Bond Order, Bond Length and Bond Strength

11A Easy

Rank C–C, C=C and C≡C in order of increasing bond length and increasing bond strength.

11B Medium

The average C–O bond order in carbonate is $\frac{4}{3}$. Explain how this accounts for its equal C–O bond lengths.

11C Hard

Rank the expected N–O bond lengths in NO_2^+ , NO_2^- and NO_3^- . Justify your answer by determining the average N–O bond order in each species.

12. Bond Polarity and Reactivity

12A Easy

Explain why HCl contains a polar bond whereas Cl_2 does not.

12B Medium

Use bond polarity to explain why nucleophiles commonly attack the carbon atom of a carbonyl group rather than its oxygen atom.

12C Hard

Acyl chlorides are generally more reactive toward nucleophilic attack than amides. Explain this using both inductive effects and electron delocalisation.

C. Molecular Geometry

13. Electron-Domain Repulsion

13A Easy

Use electron-domain repulsion to predict the shapes of CH_4, NH_3 and H_2O.

13B Medium

Predict the shape of SF_4 and explain the preferred position of its lone pair.

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13C Hard

Predict and compare the shapes of ClF_3, XeF_2 and XeF_4, showing how the electron domains arrange themselves before lone pairs are omitted from the molecular shape.

14. Bonding Pairs Versus Lone Pairs

14A Easy

State the number of bonding pairs and lone pairs around nitrogen in NH_3.

14B Medium

SO_2 has three electron domains around sulfur but is not trigonal planar as a molecular shape. Explain.

14C Hard

Predict the shape of I_3^- . Explain why three lone pairs occupy equatorial positions in the underlying trigonal-bipyramidal electron-domain arrangement.

15. Relative Strengths of Electron-Pair Repulsions

15A Easy

State the order of repulsion strength among lone pair–lone pair, lone pair–bond pair and bond pair–bond pair interactions.

15B Medium

Explain why the H–N–H angle in NH_3 is smaller than the H–C–H angle in CH_4.

15C Hard

The H–O–H angle in water is approximately 104.5°, while the F–O–F angle in OF_2 is slightly smaller. Suggest why replacing H with the more electronegative F changes the bond angle.

16. Shapes and Approximate Bond Angles

16A Easy

State the shapes and approximate bond angles of $BeCl_2$, BF_3, CH_4, NH_3 and H_2O.

16B Medium

Predict the shapes and characteristic bond angles in PCl_5 and SF_6.

16C Hard

Predict the molecular shape and approximate bond angles of BrF_5. Explain any departures from ideal angles.

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17. Effects of Multiple Bonds on Geometry

17A Easy

Why is each carbon atom in ethene approximately trigonal planar?

17B Medium

In methanal, the O–C–H angles are slightly greater than the H–C–H angle. Explain why the C=O electron domain causes this distortion.

17C Hard

Compare the geometries of NO_2^+ , NO_2^- and NO_3^- . Account for their different bond angles using electron-domain repulsion and lone pairs.

18. Electron-Pair Geometry Versus Molecular Geometry

18A Easy

State the electron-pair geometry and molecular geometry of NH_3.

18B Medium

State both geometries for SF_4 and explain why the names are different.

18C Hard

$XeOF_4$ contains four Xe–F bonds, one Xe=O bond and one lone pair around xenon. Determine its electron-pair geometry and molecular geometry.

D. Molecular Polarity

19. Electronegativity and Bond Dipoles

19A Easy

Indicate the direction of the bond dipole in H–F, C–Cl and N–H.

19B Medium

Explain why the C–F bond is more polar than the C–Cl bond even though chlorine is also highly electronegative.

19C Hard

A bond has length 1.30×10^{-10} m and dipole moment 1.50 D. Estimate the magnitude of the separated charge as a fraction of the elementary charge. Use $1\ \text{D} = 3.336 \times 10^{-30}$ C m.

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20. Vector Cancellation of Bond Dipoles

20A Easy

Explain why CO_2 is non-polar whereas H_2O is polar.

20B Medium

BF_3 and NH_3 both contain three polar bonds. Explain why BF_3 is non-polar but NH_3 is polar.

20C Hard

The two identical bond dipoles in an idealised bent molecule each have magnitude 1.50 D and form an angle of 119°. Calculate the resultant molecular dipole moment.

21. Polar Bonds in Non-Polar Molecules

21A Easy

Give one molecule that contains polar bonds but has no permanent molecular dipole.

21B Medium

Explain why CCl_4 is non-polar but $CHCl_3$ is polar.

21C Hard

Compare the molecular polarities of cis- and trans-1,2-dichloroethene using vector diagrams of their bond dipoles.

22. Molecular Symmetry

22A Easy

Explain how symmetry causes the bond dipoles in BF_3 to cancel.

22B Medium

Predict whether CF_4, CF_3Cl and CF_2Cl_2 are polar.

22C Hard

The three isomers of dichlorobenzene have the same bonds but different arrangements. Predict which isomer should have the smallest molecular dipole moment and justify your answer geometrically.

23. Effect of Lone Pairs on Polarity

23A Easy

Explain why NH_3 has a permanent dipole.

23B Medium

Compare the polarity of BF_3 and NF_3, including the influence of the lone pair on nitrogen.

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23C Hard

XeF_2 is non-polar whereas ClF_3 is polar, even though both have trigonal-bipyramidal electron-domain arrangements. Explain.

E. Intermolecular Forces

24. Instantaneous Dipole–Induced Dipole Forces

24A Easy

Explain how London dispersion forces arise in a non-polar atom or molecule.

24B Medium

Explain why xenon has a higher boiling point than argon.

24C Hard

F_2, Cl_2, Br_2 and I_2 have the same general molecular shape. Explain their progression from gases to liquid to solid at room temperature.

25. Permanent Dipole–Permanent Dipole Forces

25A Easy

What molecular property is required for permanent dipole–dipole attraction?

25B Medium

Propanone has a substantially higher boiling point than propane despite having a similar relative molecular mass. Explain.

25C Hard

Cis-1,2-dichloroethene has a higher boiling point than its trans isomer, but the trans isomer may pack more efficiently in a crystal. Predict how their boiling points and melting points might differ.

26. Hydrogen Bonding

26A Easy

State the structural requirements for conventional hydrogen bonding between molecules.

26B Medium

Ethanol and dimethyl ether have the same molecular formula. Explain why ethanol has the higher boiling point.

26C Hard

Ortho-nitrophenol can form an intramolecular hydrogen bond, whereas para-nitrophenol more readily forms intermolecular hydrogen bonds. Predict their relative boiling points and water solubilities.

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27. Polarisability

27A Easy

Define polarisability.

27B Medium

Explain why I_2 is more polarisable than Cl_2.

27C Hard

A large molecule may possess no permanent dipole yet have a higher boiling point than a small polar molecule. Explain how this is possible without claiming that London forces are intrinsically a different kind of electrostatic interaction.

28. Molecular Size, Surface Area and Shape

28A Easy

Explain why hexane has a higher boiling point than butane.

28B Medium

Rank pentane, 2-methylbutane and 2,2-dimethylpropane by boiling point.

28C Hard

Branching normally lowers the boiling point of an alkane but can sometimes increase its melting point. Explain why the effects on boiling and melting need not follow the same trend.

29. Intermolecular Versus Intramolecular Bonding

29A Easy

When water boils, are O–H covalent bonds broken? Explain.

29B Medium

Distinguish the energy changes involved in melting NaCl from those involved in decomposing NaCl into neutral sodium and chlorine atoms.

29C Hard

A student says that diamond has a high boiling point because it has “very strong intermolecular forces”. Correct this explanation and distinguish clearly between a molecular substance and a network solid.

30. Solubility and “Like Dissolves Like”

30A Easy

Explain why hexane and water form separate layers.

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30B Medium

Ethanol mixes with both water and hexane to some extent. Explain this using different parts of the ethanol molecule.

30C Hard

Compare the likely water solubilities of benzene, phenol and benzene-1,2-diol. Discuss solute–solvent attractions, disruption of existing interactions and the limitations of simply counting hydrogen bonds.

F. Structures of Solids

31. Ionic Lattices

31A Easy

Explain why solid NaCl does not conduct electricity but molten NaCl does.

31B Medium

MgO has a much higher melting point than NaCl. Explain using ionic charge, ionic radius and lattice attraction.

31C Hard

A highly charged ionic solid can have a very large lattice enthalpy but still dissolve poorly in water. Explain why lattice enthalpy alone cannot determine solubility.

32. Metallic Lattices

32A Easy

Use metallic bonding to explain electrical conductivity and malleability.

32B Medium

Magnesium generally has a higher melting point than sodium. Explain using the charge and size of the metal ions and the number of delocalised electrons.

32C Hard

Pure copper is softer than brass. Explain how differently sized atoms in an alloy interfere with movement of layers while retaining metallic conductivity.

33. Simple Molecular Solids

33A Easy

Explain why iodine forms a solid that melts at a relatively low temperature and does not conduct electricity.

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33B Medium

Compare the structures and melting behaviour of solid CO_2 and SiO_2.

33C Hard

Buckminsterfullerene, C_{60}, is composed only of carbon but has a much lower melting or sublimation temperature than diamond. Explain.

34. Giant Covalent Structures

34A Easy

Explain why diamond is hard and has a very high melting temperature.

34B Medium

SiO_2 and CO_2 contain bonds between elements from the same groups but form very different solids. Explain why.

34C Hard

Silicon and diamond both possess extended covalent structures, but silicon is a semiconductor while diamond is normally an electrical insulator. Suggest a bonding-based explanation.

35. Hydrogen-Bonded Solids

35A Easy

Describe the role of hydrogen bonding in holding an ice crystal together.

35B Medium

Explain why ice forms an open lattice and is less dense than liquid water.

35C Hard

Urea forms a hydrogen-bonded molecular solid with a relatively high melting point. Explain why it is nevertheless classified as a molecular solid rather than a giant covalent structure.

36. Structure–Property Relationships

36A Easy

For NaCl, copper, iodine and diamond, state the dominant bonding or attractive force and whether each conducts electricity as a solid.

36B Medium

An unknown solid has a high melting point, is brittle, dissolves in water and conducts only when molten or dissolved. Deduce its likely structure.

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36C Hard

Three unknown solids have the following properties: P has a low melting point, is volatile and non-conducting; Q has a very high melting point, is hard and conducts as a solid; R has a high melting point, is brittle and conducts only when molten. Propose possible structures for P, Q and R, and identify what additional evidence would remove ambiguity.

37. Graphite, Diamond and Unfamiliar Allotropes

37A Easy

Compare the bonding and electrical conductivity of graphite and diamond.

37B Medium

Explain why graphite is soft even though the covalent bonds within each carbon layer are strong.

37C Hard

An unfamiliar carbon allotrope consists of covalently bonded two-dimensional sheets in which every carbon forms three σ bonds and contributes one electron to a delocalised π system. Predict its conductivity, mechanical strength within a sheet and ease of separating the sheets.

G. Integrated Problem Types

38. Boiling-Point Ranking

38A Easy

Rank CH_4, NH_3 and H_2O by boiling point and explain the main intermolecular forces present.

38B Medium

Rank butane, propanone, propan-1-ol and ethanoic acid by expected boiling point.

38C Hard

Rank the boiling points of pentane, 2-methylbutane, butan-1-ol, ethoxyethane and butanoic acid. Explain every adjacent comparison rather than invoking only relative molecular mass.

39. Structure from Physical-Property Data

39A Easy

A substance has a low melting point, does not conduct in any state and dissolves in hexane. Suggest its likely structure.

39B Medium

A solid is soft, electrically conducting and composed entirely of carbon. Identify a likely allotrope and explain its properties.

39C Hard

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A colourless solid melts at 780°C, dissolves in water, does not conduct as a solid and produces a conducting solution. Deduce what can and cannot be concluded about its bonding.

40. Comparing Isomers

40A Easy

Which has the higher boiling point: pentane or 2,2-dimethylpropane?

40B Medium

Explain why straight-chain alkanes generally have higher boiling points than their more highly branched isomers.

40C Hard

One highly symmetrical branched isomer has a lower boiling point but a higher melting point than a less symmetrical straight-chain isomer. Explain both trends.

41. Identifying All Intermolecular Forces

41A Easy

Identify all intermolecular forces between HCl molecules.

41B Medium

Identify all intermolecular forces present in pure ethanol. Explain why London forces should not be omitted.

41C Hard

Identify and compare the interactions present in pure ethanoic acid, ethanoic acid dissolved in water and gaseous ethanoic acid dimers.

42. Predicting Polarity from Unfamiliar Structures

42A Easy

Predict whether SO_3 is polar.

42B Medium

Predict whether SF_4 and XeF_4 are polar.

42C Hard

A molecule AX_4Y has trigonal-bipyramidal electron geometry, with Y occupying either an axial or equatorial site. Determine which arrangement is likely to possess the larger molecular dipole and what further information is needed.

43. Anomalous Properties of Water and Ice

43A Easy

Explain why water has a much higher boiling point than H_2S.

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43B Medium

Explain why ice floats on liquid water.

43C Hard

Water reaches maximum density near 4°C, rather than at its freezing point. Propose a molecular explanation involving competition between thermal contraction and hydrogen-bonded structural organisation.

44. Bond-Length and Bond-Energy Comparisons

44A Easy

Compare the bond lengths and strengths of C–C, C=C and C≡C.

44B Medium

Explain why the C–O bonds in carbonate are shorter than a normal C–O single bond but longer than a C=O double bond.

44C Hard

Arrange the N–O bonds in NO^+ , NO_2^- and NO_3^- by expected length. State clearly any assumptions used.

45. Testing Proposed Lewis Structures and Shapes

45A Easy

A student proposes that CO_2 is bent because each oxygen has two lone pairs. Identify the error.

45B Medium

A student proposes a trigonal-planar shape for NH_3. Explain why this is not the molecular geometry.

45C Hard

A proposed structure for XeF_4 is tetrahedral because xenon forms four bonds. Use electron counting and electron-domain geometry to evaluate the proposal.

Olympiad-Style Extensions

The following problems focus on critical thinking rather than specialised prior knowledge.

Extension 1: Counting Particles in a Unit Cell

A crystalline solid contains two elements, A and B.

- A atoms occupy all eight corners of a cubic unit cell.
- B atoms occupy the centres of all six faces.

1(a) Effective number of A atoms

Calculate the effective number of A atoms in one unit cell.

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1(b) Effective number of B atoms

Calculate the effective number of B atoms in one unit cell.

1(c) Empirical formula

Determine the empirical formula of the solid.

1(d) Recognising a cubic structure

Suppose all A and B atoms were instead atoms of the same element X. How many X atoms would be present effectively in the unit cell, and what type of cubic structure would this represent?

1(e) Coordination beyond one unit cell

A student says that the corner A atoms are bonded only to the three B atoms on the three visible faces meeting at that corner. Explain why this conclusion is incorrect.

Extension 2: Identifying a Unit Cell Using Density

A metal X forms a cubic crystal. Its unit-cell edge length is

$$a = 400 \text{ pm}.$$

The metal has molar mass

$$M = 63.5 \text{ g mol}^{-1}$$

and measured density

$$\rho = 6.59 \text{ g cm}^{-3}.$$

Possible structures are:

Structure	Effective atoms per unit cell
Simple cubic	1
Body-centred cubic	2
Face-centred cubic	4

Use

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}.$$

2(a) Converting the unit-cell edge length

Convert the unit-cell edge length into centimetres.

2(b) Volume of one unit cell

Calculate the volume of one unit cell.

2(c) Mass of one unit cell

Calculate the mass of one unit cell.

2(d) Effective number of atoms

Calculate the effective number of X atoms in each unit cell.

2(e) Identifying the crystal structure

Identify the crystal structure.

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2(f) Why a small whole number is expected

Explain why the calculated number of atoms should be close to a small whole number.

Extension 3: Interpreting Bond Potential-Energy Curves

The graph represents the potential energy of two pairs of bonded atoms, P and Q, as a function of internuclear distance. The potential energy is defined to approach zero as the atoms become infinitely separated.

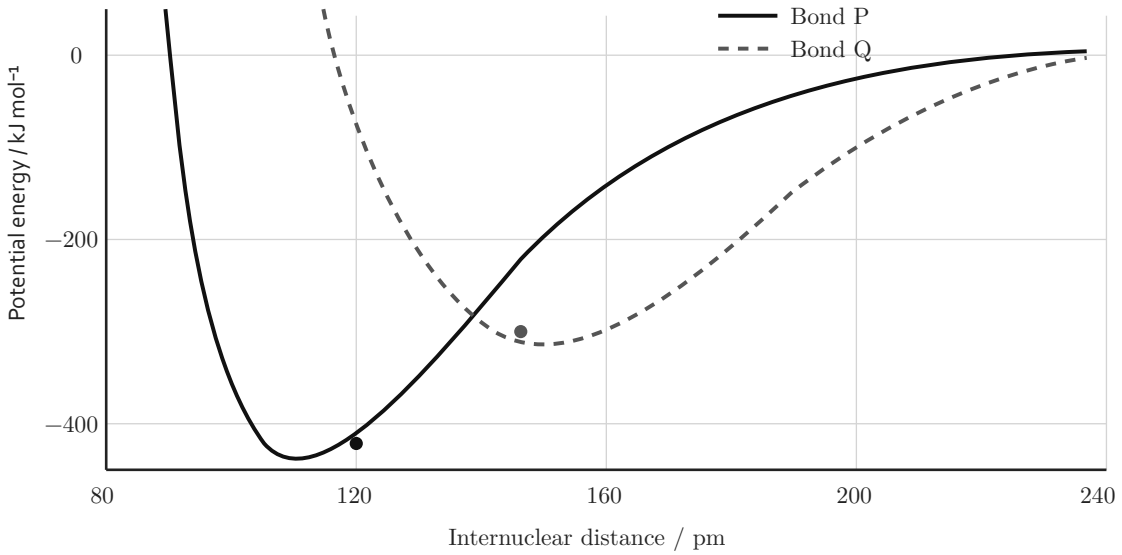


Figure 1: Potential-energy curves for bonds P and Q.

Bond	Distance at energy minimum	Minimum potential energy
P	120 pm	-440 kJ mol^{-1}
Q	155 pm	-250 kJ mol^{-1}

3(a) Meaning of the energy minimum

What does the distance at the minimum of each curve represent?

3(b) Comparing bond lengths

Which bond is shorter?

3(c) Comparing bond strengths

Which bond is stronger?

3(d) Short-range repulsion

Explain why the potential energy rises very steeply when the atoms are brought much closer together than the equilibrium bond length.

3(e) Large-separation limit

At a distance much greater than the equilibrium bond length, why does the potential energy approach zero?

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3(f) Stretching vibrational frequency

Which bond would probably have the higher stretching vibrational frequency, assuming the bonded atoms have similar masses?

3(g) Equilibrium distance and vibration

A student concludes that the atoms in bond P are stationary at exactly 120 pm. Explain why this is not strictly correct.

Extension 4: Interpreting a Boiling-Point Graph

The approximate boiling points of several hydrides are shown below.

Molecule	Relative molecular mass	Boiling point / K
CH_4	16	112
SiH_4	32	161
GeH_4	77	185
SnH_4	123	221
NH_3	17	240
H_2O	18	373

Molecule	Relative molecular mass	Boiling point / K
HF	20	293

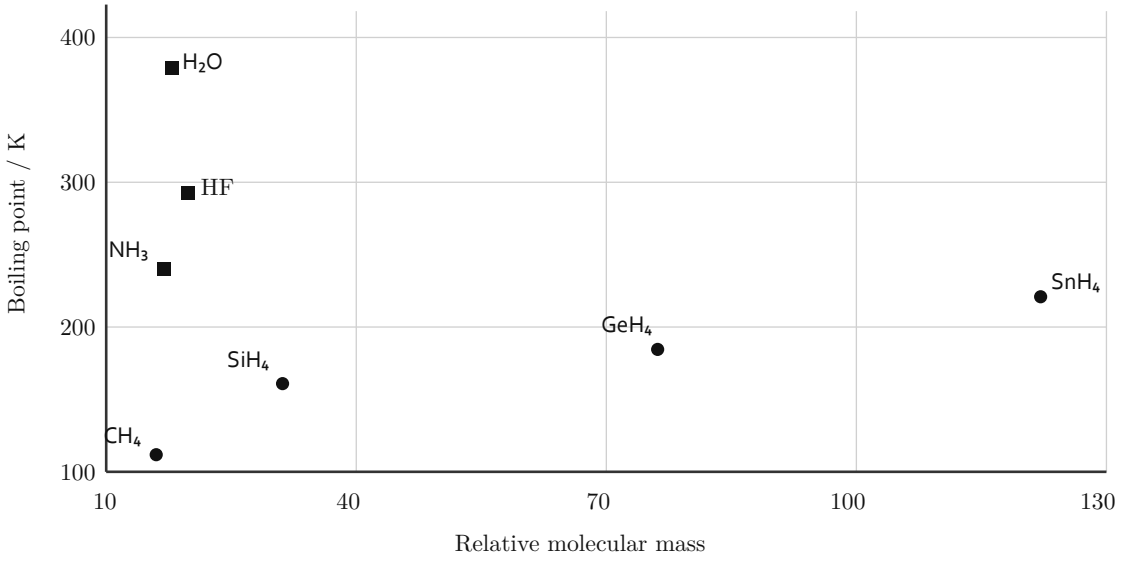


Figure 2: Boiling point against relative molecular mass for selected hydrides.

4(a) Trend among Group 14 hydrides

Describe the trend shown by CH_4, SiH_4, GeH_4, and SnH_4.

4(b) Why "larger molecules" is incomplete

Why is it incomplete to say only that "larger molecules have stronger intermolecular forces"?

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4(c) Major outliers

Identify the three major outliers from the trend shown by the Group 14 hydrides.

4(d) Hydrogen bonding and boiling point

Explain why NH_3, H_2O, and HF lie above the general trend.

4(e) Why water boils above HF

H_2O has a higher boiling point than HF even though the H–F bond is more polar than the O–H bond. Explain.

4(f) Predicting the boiling point of hydrogen sulfide

Predict whether the boiling point of H_2S would be closer to that of H_2O or to the smooth trend shown by the Group 14 hydrides.

4(g) Do London forces disappear?

A student states, “NH3 has hydrogen bonding, so London forces are absent.” Evaluate this statement.

Extension 5: Inferring Structure from Conductivity Data

The electrical conductivities of four substances are measured in different states.

Substance	Solid	Molten	Dissolved in water
W	Very low	High	High
X	High	High	Insoluble
Y	Very low	Very low	Very low
Z	High	—	Insoluble

Further observations are:

- W is brittle and has a high melting point.
- X is malleable.
- Y is volatile and has a low melting point.
- Z is very hard and has an extremely high melting point.

5(a) Structure of W

Deduce the likely structure of W.

5(b) Structure of X

Deduce the likely structure of X.

5(c) Structure of Y

Deduce the likely structure of Y.

5(d) A possible structure for Z

Deduce a possible structure for Z.

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5(e) Limits of conductivity data

Explain why conductivity data alone may not uniquely identify a structure.

End of question set

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